

Power Meter

Brief

A custom-designed PCB instrument capable of measuring 13 AC and DC electrical quantities in real time, including RMS voltage and current, frequency, phase difference, real/reactive/apparent power, and power factor. Built around an ATmega328P microcontroller, the device features a 128×64 LCD display with configurable measurement views, five push buttons for navigation, automatic high/low current range switching, and a real-time clock for timestamped logging.

The hardware was designed entirely in Altium Designer, with signal conditioning circuits, op-amp stages and voltage dividers, simulated in Multisim to ensure all ADC inputs remained within the safe 0–3.3V range. The measurement firmware, written in C, derived all 13 quantities from raw ADC readings through a combination of RMS sampling, zero-crossing detection for frequency, and a Discrete Fourier Transform approach for phase difference and power factor. A companion Python PC application streamed all measurements live over UART, with selectable real-time graphing, RTC-timestamped data points, and under-limit flagging for out-of-range signals. The final product was fabricated as a custom PCB and successfully demonstrated the majority of project specification requirements.

Demo Videos:

- GUI Demo (PC Application): <https://youtu.be/u9A9L5Fke7U>
- Breadboard Demo (Embedded Firmware & LCD): <https://youtu.be/5EAJGChlrte>

Final PCB:

